

About the Learner:

I started at Northeastern studying computer science and physics without knowing how they'd connect. Computer science came from high school aptitude. Physics came from genuine interest. After one biology semester convinced me it wasn't for me, I committed to both.

My Via Separations co-op taught me how to build systems under real constraints. Engineering a warehouse management system for 50+ users meant thinking about data models that won't break at scale, authentication flows, and what happens when production systems fail. In my role as technical lead at Generate Product Development, I built StrideTrack, a wearable biomechanics visualization platform, and Cortex, an unstructured manufacturing document pattern recognition pipeline, learning how to translate vague client requirements into working infrastructure. These roles taught me valuable skills, but they also revealed something: traditional software engineering can feel repetitive. You're building better versions of existing systems, solving problems that have known solution patterns. Research offers something different. The problems are genuinely unsolved, and the rabbit holes you go down while figuring them out are intellectually stimulating in a way that standard web applications aren't.

Graph neural networks became my entry point. These are machine learning architectures that operate directly on graph-structured data like molecular or crystal structures. Professor Yan presented me with an opportunity to work on defect formation energy prediction, and I got interested in how these models predict materials properties from atomic arrangements without hand-crafted features. What draws me broadly is how new ML technology can address concrete scientific bottlenecks like quantum computing, better batteries, or next-generation semiconductors. The computational side is interesting, but the application is what matters.

I'm confident in implementing ML systems. I can read a paper, code the method, and debug when things break. I understand graph neural networks and have solid foundations in neural architectures generally. Where I'm learning is physics. For example, I know what density functional theory (DFT) calculations do conceptually, but I need hands-on experience running them, interpreting electronic structures, and connecting computational outputs to actual physical behavior. Reinforcement learning sits in between. I am taking the course and implemented Monte Carlo Policy Gradient for option hedging, but I haven't applied it to generative models.

I'll meet weekly with Professor Yan, showing training curves, validation results, and code for technical discussions. My primary learning goals are to:

1. understand the physics deeply enough to recognize when results make physical sense
2. build a working reinforcement learning pipeline from generation through validation
3. explain this methodology clearly to non-specialists through my RISE presentation

Purpose and Significance:

Quantum computers require 15 millikelvin cooling with \$500,000+ dilution refrigerators, limiting them to specialized labs. Materials that work at accessible temperatures (above 77 K) would change this, but they must satisfy three criteria simultaneously: topological protection (quantum states protected by symmetry), superconducting behavior at elevated temperatures, and synthesizability.

Most computational materials discovery optimizes for one property. Microsoft's MatterGen targets thermodynamic stability. Other approaches target specific band gaps or magnetic properties. The critical gap is synthesizability. Generative models can propose thousands of topologically interesting materials, but if they require precursors that don't exist or synthesis conditions only achievable in specialized equipment, they're computationally interesting but practically useless.

Reinforcement learning is a machine learning approach where an AI agent learns through trial and error guided by reward signals. The question is whether multi-objective RL can discover materials balancing all three criteria simultaneously. The Topological Quantum Chemistry database contains 38,184 classified materials but only approximately 12 confirmed topological superconductors, none above 77 K. Recent RL work shows 378 times better data efficiency than traditional methods, but no existing work targets the joint optimization problem for quantum materials. Success would establish a methodology for multi-constraint discovery applicable to other problems and potentially identify materials enabling scalable quantum computing.

Objectives and Methods:

This project will implement a multi-objective RL pipeline, generate candidate structures, identify several stable topological materials through DFT validation, and measure efficiency gains. I'll download Topological Quantum Chemistry (38,184 materials), JARVIS-DFT (1,868 topological materials), and Materials Project databases during weeks 1-2 (June 8-21, 30 hours/week). I'll set up pre-trained models: DiffCSP (Microsoft's crystal generator), ALIGNN (NIST's Atomistic Line Graph Neural Network for property prediction), M3GNet (Materials Project's stability evaluator). Backup models are available if validation fails. I'll generate baseline structures to establish expected hit rates.

Weeks 3-5 (June 22-July 12, 10-25 hours/week) focus on implementing the MatInvent RL framework. The multi-objective reward combines topological probability, density of states at Fermi energy (superconducting proxy), formation energy (stability/synthesizability), and composition penalties favoring light elements. Proximal Policy Optimization (PPO) generates candidate structures with reward weights tuned to balance objectives rather than sacrifice one for another. I'll periodically validate top-ranked structures with DFT and retrain predictors as needed.

During weeks 6-8 (July 13-August 2, 10-15 hours/week), I'll run VASP (Vienna Ab initio Simulation Package) calculations on the top candidates: structure relaxation, electronic structure with spin-orbit coupling, topological invariant checks. For promising materials, I'll compute phonon dispersions via Quantum ESPRESSO to confirm stability and estimate superconducting temperatures. I'll use Professor Yan's HPC cluster primarily with Discovery cluster as backup. Weeks 9-10 (August 3-16, 15 hours/week) focus on extracting Pareto fronts, creating visualizations, and benchmarking against unconditional generation. The timeline accommodates my co-op starting July 6 and allows long-running calculations during reduced-hour weeks.

Potential challenges include prediction accuracy degrading for novel structures (switch to simpler proxies), convergence to known materials (add novelty rewards), queue delays (parallel submission), and phonon convergence failures (prioritize small unit cells).

Outcomes, Evaluation, and Dissemination:

Three checkpoints structure evaluation: Week 2 confirms models match published performance, Week 5 measures RL improvement over baseline, Week 8 validates discovery of several stable topological materials. I'll benchmark against unconditional generation, random database sampling, and literature values. Outcomes include a validated material database, an open-source RL pipeline, and composition-structure insights.

I'll present at RISE, work with Professor Yan to submit to the Materials Research Society, release code on GitHub, and contribute structures to the Materials Project. If results warrant, we'll discuss journal manuscript preparation.